KCE PATHWAYS · GRADE 3 · UNIT 3: MARKET DAYS · WEEK 11 · MATH

Harvest Day at the Citrus Grove — Teacher Lesson Plan & Answer Key

Learning objectives: Students fluently multiply with factors 6, 7, 8 and 9 and use the turn-around (commutative) property, the break-apart (distributive) property, and the identity and zero properties to find and check products. They solve one- and two-step multiplication word problems and explain or correct a strategy.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: student types first and last name so results go to KingCretot Experience.
2. Warm-Up (Unit 2 review, 15 pts): equal groups, skip-counting, division and fact families, two-step problems. If a student misses two of three, review the Week 6–9 ideas before moving on.
3. Picture It: Harvest Day at the Citrus Grove. Introduces Rosa's grove near Arcadia and Florida citrus facts. Point out that rows of trees are arrays.
4. Word Wrangler: 6 vocab cards (factor, product, turn-around, break-apart, identity, zero properties). Mention the formal names commutative and distributive.
5. Pencil Break (Pre): student completes the Pre side of the printed sheet before any teaching. Do not help yet; it shows what they already know.
6. Turn-Around Facts: 6s and 7s (15 pts). Tip: have the student turn a real tray or paper array sideways to see the total does not change.
7. The Break-Apart Trick (15 pts). Tip: encourage breaking a factor into 5 + something, since 5s are known facts. Ask the student to say both partial products aloud before adding.
8. Patterns in the 8s and 9s, plus 1 and 0 (15 pts). Tip: for 9s, have the student check that the digits add to 9; for 8s, double the 4s fact.
9. Bonus Practice: Packing Crates (XP only).
10. Match-Up Round (10 pts, 2 per first-try match).
11. Fix the Sentence (10 pts): typed answers are checked exactly (capital letter, end mark, list commas). Both serial-comma and no-serial-comma list forms are accepted.
12. Speed Arena and Keyboard Corral (XP only).
13. Boss Battle: The Big Harvest (20 pts): two-step problem, which-explanation-is-correct, word problem, find-the-mistake.
14. Pencil Break (Post): student completes the Post side of the sheet and the star challenge if time allows.
15. Results: score of 85 or higher shows a custom Harvest Hero congratulations with confetti; below 85 shows encouragement and a fresh question set on the next try (sets rotate 1, 2, 3).

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: UNIT 2 REVIEW

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Mia sees 4 rows of jars with 5 jars in each row. How many jars?	**20**	4 rows of 5 is 4 × 5 = 20 jars.
w2_s0	Set 1: If 6 × 4 = 24, then 24 ÷ 4 = ___	**6**	24 ÷ 4 = 6 because 6 × 4 = 24. Multiplication and division undo each other.
w3_s0	Set 1: Theo had 3 boxes of 5 maps. He sold 4 maps. How many maps are left?	**11**	Step 1: 3 × 5 = 15 maps. Step 2: 15 − 4 = 11 maps left.
w1_s1	Set 2: Skip-count by 5s: 5, 10, 15, 20, ___	**25**	Skip-counting by 5s adds 5 each jump: 20 + 5 = 25.
w2_s1	Set 2: Which is the same as 3 + 3 + 3 + 3?	**4 × 3**	Four groups of 3 is 4 × 3 = 12.
w3_s1	Set 2: 20 apples are shared equally among 5 baskets. How many apples go in each basket?	**4**	20 ÷ 5 = 4 apples in each basket.
w1_s2	Set 3: Grandpa sells candles for $10 each. Mia buys 3 candles. How many dollars does she pay?	**30**	10 × 3 = 30 dollars.
w2_s2	Set 3: Which fact is in the same fact family as 3 × 7 = 21?	**21 ÷ 7 = 3**	21 ÷ 7 = 3 uses the same three numbers, 3, 7 and 21.
w3_s2	Set 3: Mia had 2 bags of 10 stickers. She gave away 5 stickers. How many are left?	**15**	Step 1: 2 × 10 = 20 stickers. Step 2: 20 − 5 = 15 left.

TURN-AROUND FACTS: 6S AND 7S

ID	Question	Answer	Why (explain to student)
t1a_s0	Set 1: Rosa knows 6 × 8 = 48. What is 8 × 6?	**48**	Turn-around facts have the same product: 8 × 6 = 48.
t1b_s0	Set 1: A block of the grove has 7 rows with 6 trees in each row. How many trees?	**42**	7 × 6 = 42 trees.
t1c_s0	Set 1: Which explanation is correct? Theo says 9 × 6 and 6 × 9 have different products because the numbers are in a different order. Mia says they have the same product because of the turn-around property.	**Mia is correct**	Switching the order of the factors does not change the product: 9 × 6 = 6 × 9 = 54.
t1a_s1	Set 2: Which fact is the turn-around of 7 × 9 = 63?	**9 × 7 = 63**	A turn-around fact switches the order of the factors: 9 × 7 = 63.
t1b_s1	Set 2: A picker fills 6 bags of oranges every hour. How many bags does she fill in 6 hours?	**36**	6 × 6 = 36 bags.
t1c_s1	Set 2: Find the mistake. Rosa wrote: 7 × 6 = 42 and 6 × 7 = 48. What is wrong?	**6 × 7 = 48 is wrong. It should be 42.**	Turn-around facts are equal, so 6 × 7 must also be 42, not 48.
t1a_s2	Set 3: 8 × 7 = 56. Which other fact must also equal 56?	**7 × 8**	7 × 8 is the turn-around of 8 × 7, so it is also 56.
t1b_s2	Set 3: The juice stand sells cups of orange juice for $7 each. Mia sells 6 cups. How many dollars does she collect?	**42**	6 × 7 = 42 dollars.

ID	Question	Answer	Why (explain to student)
t1c_s2	Set 3: Which explanation is correct? Why is 6 × 9 the same as 9 × 6?	**Turning an array on its side does not change how many are in it.**	An array of 6 rows of 9 turned on its side is 9 rows of 6. It has the same 54 items.

THE BREAK-APART TRICK

ID	Question	Answer	Why (explain to student)
t2a_s0	Set 1: Use the break-apart trick. 7 × 8 = 7 × 5 + 7 × 3 = 35 + 21 = ___	**56**	35 + 21 = 56, so 7 × 8 = 56.
t2b_s0	Set 1: Which is a correct way to break apart 6 × 7?	**6 × 5 + 6 × 2**	7 = 5 + 2, so 6 × 7 = 6 × 5 + 6 × 2 = 30 + 12 = 42.
t2c_s0	Set 1: Find the mistake. Theo wrote: 8 × 5 = 40 and 8 × 1 = 8, so 8 × 6 = 40 + 1 = 41. What went wrong?	**He added 1 instead of 8. It should be 40 + 8 = 48.**	The two parts are 8 × 5 = 40 and 8 × 1 = 8. Add the products: 40 + 8 = 48.
t2a_s1	Set 2: Use the break-apart trick. 6 × 8 = 6 × 5 + 6 × 3 = 30 + 18 = ___	**48**	30 + 18 = 48, so 6 × 8 = 48.
t2b_s1	Set 2: A truck holds 9 rows of 7 bins. Break 7 into 5 and 2: 9 × 5 = 45 and 9 × 2 = 18. How many bins in all?	**63**	45 + 18 = 63 bins, so 9 × 7 = 63.
t2c_s1	Set 2: Which explanation is correct? Rosa says 7 × 9 = 7 × 5 + 7 × 4.	**She is right, because 5 + 4 = 9, so the two parts make 7 × 9.**	9 = 5 + 4, so 7 × 9 = 35 + 28 = 63.
t2a_s2	Set 3: Use the break-apart trick. 8 × 7 = 8 × 5 + 8 × 2 = 40 + 16 = ___	**56**	40 + 16 = 56, so 8 × 7 = 56.
t2b_s2	Set 3: To find 9 × 6, Mia breaks 6 into 5 + 1. Which shows her work?	**9 × 5 + 9 × 1 = 45 + 9**	Multiply 9 by each part: 45 + 9 = 54, so 9 × 6 = 54.
t2c_s2	Set 3: Find the mistake. Theo wrote: 7 × 7 = 7 × 5 + 7 × 2 = 35 + 12 = 47. Which part is wrong?	**7 × 2 is 14, not 12, so the answer is 49.**	7 × 2 = 14. Then 35 + 14 = 49, so 7 × 7 = 49.

PATTERNS IN THE 8S AND 9S, PLUS 1 AND 0

ID	Question	Answer	Why (explain to student)
t3a_s0	Set 1: Use the 9s pattern. The tens digit is one less than 7. 9 × 7 = ___	**63**	The tens digit is 6, and 6 + 3 = 9, so 9 × 7 = 63.
t3b_s0	Set 1: Rosa has 8 empty crates with 0 oranges in each. How many oranges are in the crates?	**0**	8 × 0 = 0. Any number times 0 is 0.
t3c_s0	Set 1: Which explanation of the 9s pattern is correct for 9 × 8 = 72?	**The tens digit 7 is one less than 8, and 7 + 2 = 9.**	In the 9s, the tens digit is one less than the other factor, and the digits add to 9.
t3a_s1	Set 2: 8 × 6 is double 4 × 6. Since 4 × 6 = 24, 8 × 6 = ___	**48**	Double 24 is 48, so 8 × 6 = 48.
t3b_s1	Set 2: Mia packed 1 crate with 9 oranges. How many oranges? 9 × 1 = ___	**9**	Any number times 1 is that same number: 9 × 1 = 9.
t3c_s1	Set 2: Find the mistake. Theo wrote 9 × 4 = 63. What is wrong?	**The tens digit should be one less than 4, so the answer is 36.**	The tens digit is 3, and 3 + 6 = 9, so 9 × 4 = 36. Theo switched the digits.

ID	Question	Answer	Why (explain to student)
t3a_s2	Set 3: Use the 9s pattern. 9 × 9 = ___	**81**	The tens digit is one less than 9, which is 8, and 8 + 1 = 9, so 9 × 9 = 81.
t3b_s2	Set 3: Which fact is true?	**7 × 0 = 0**	Any number times 0 is 0, and any number times 1 is itself. So 7 × 0 = 0 is true.
t3c_s2	Set 3: Two steps! The juice stand has 8 bags with 8 oranges in each bag. Mia uses 4 oranges to make juice. How many oranges are left?	**60**	Step 1: 8 × 8 = 64 oranges. Step 2: 64 − 4 = 60 left.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 6 × 8	**48**	2 points if matched on the first try.
match_s0_2	Set 1: Match 7 × 9	**63**	2 points if matched on the first try.
match_s0_3	Set 1: Match 9 × 6	**54**	2 points if matched on the first try.
match_s0_4	Set 1: Match 8 × 1	**8**	2 points if matched on the first try.
match_s0_5	Set 1: Match 7 × 0	**0**	2 points if matched on the first try.
match_s1_1	Set 2: Match 8 × 7 = 7 × 8	**turn-around property**	2 points if matched on the first try.
match_s1_2	Set 2: Match 7 × 8 = 7 × 5 + 7 × 3	**break-apart property**	2 points if matched on the first try.
match_s1_3	Set 2: Match 9 × 1 = 9	**identity property**	2 points if matched on the first try.
match_s1_4	Set 2: Match 6 × 0 = 0	**zero property**	2 points if matched on the first try.
match_s1_5	Set 2: Match 9 × 8	**72**	2 points if matched on the first try.
match_s2_1	Set 3: Match 6 × 6	**36**	2 points if matched on the first try.
match_s2_2	Set 3: Match 7 × 8	**56**	2 points if matched on the first try.
match_s2_3	Set 3: Match 9 × 9	**81**	2 points if matched on the first try.
match_s2_4	Set 3: Match 8 × 6	**48**	2 points if matched on the first try.
match_s2_5	Set 3: Match 9 × 7	**63**	2 points if matched on the first try.

FIX THE SENTENCE

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: rosa picked 6 crates of oranges	**Rosa picked 6 crates of oranges.**	A sentence starts with a capital letter, Rosa is a name, and a telling sentence ends with a period.
fs2_s0	Set 1: Fix the sentence: we packed oranges lemons and limes	**We packed oranges, lemons, and limes. (also accepted: We packed oranges, lemons and limes.)**	Start with a capital W, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: theo counted 7 rows of trees	**Theo counted 7 rows of trees.**	Theo is a name and starts the sentence, so it gets a capital T. End the telling sentence with a period.

ID	Question	Answer	Why (explain to student)
fs2_s1	Set 2: Fix the sentence: the truck carried bins crates and bags	**The truck carried bins, crates, and bags. (also accepted: The truck carried bins, crates and bags.)**	Start with a capital T, put commas between bins, crates, and bags, and end with a period.
fs1_s2	Set 3: Fix the sentence: how many oranges are in the crate	**How many oranges are in the crate?**	Start with a capital H. This sentence asks something, so it ends with a question mark.
fs2_s2	Set 3: Fix the sentence: rosa theo and mia ate oranges	**Rosa, Theo, and Mia ate oranges. (also accepted: Rosa, Theo and Mia ate oranges.)**	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.

BOSS BATTLE: THE BIG HARVEST

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Two steps! The truck carries 7 rows of 9 bins. Rosa's family unloads 10 bins at the packing house. How many bins are still on the truck?	**53**	Step 1: 7 × 9 = 63 bins. Step 2: 63 − 10 = 53 bins still on the truck.
bb2_s0	Set 1: Which explanation is correct? Mia found 8 × 7 like this: 8 × 5 + 8 × 2 = 40 + 16 = 56. Theo found it like this: 8 + 7 = 15.	**Mia is correct.**	Mia used the break-apart trick correctly. Theo added instead of multiplying. 8 × 7 = 56.
bb3_s0	Set 1: Cups of fresh orange juice cost $6 each. A family buys 9 cups. How many dollars do they pay?	**54**	9 × 6 = 54 dollars.
bb4_s0	Set 1: Find the mistake. Rosa says 9 × 3 = 37. What is wrong?	**In the 9s the digits add to 9, so 9 × 3 = 27.**	The tens digit is one less than 3, which is 2, and 2 + 7 = 9. So 9 × 3 = 27.
bb1_s1	Set 2: Two steps! Pickers filled 6 bags with 8 oranges in each bag. Then they found 5 more oranges. How many oranges in all?	**53**	Step 1: 6 × 8 = 48 oranges. Step 2: 48 + 5 = 53 oranges in all.
bb2_s1	Set 2: Find the mistake. Theo says 6 × 9 = 54, so 9 × 6 = 56. What is wrong?	**Turn-around facts are equal, so 9 × 6 = 54.**	Switching the order of the factors does not change the product, so 9 × 6 = 54.
bb3_s1	Set 2: Each crate holds 8 oranges. How many oranges are in 9 crates?	**72**	9 × 8 = 72 oranges.
bb4_s1	Set 2: Which explanation is correct? How can 7 × 6 = 42 help you find 7 × 7?	**7 × 7 is one more group of 7, so 42 + 7 = 49.**	7 × 7 = 7 × 6 + 7 × 1 = 42 + 7 = 49.
bb1_s2	Set 3: Two steps! Rosa packs 7 crates with 9 oranges each and 1 crate with 6 oranges. How many oranges in all?	**69**	Step 1: 7 × 9 = 63 oranges. Step 2: 63 + 6 = 69 oranges in all.
bb2_s2	Set 3: Which explanation is correct? Why does 0 × 8 = 0?	**There are 0 groups, so there is nothing to count.**	Zero groups of 8 means no oranges at all, so the product is 0.
bb3_s2	Set 3: A block of the grove has 8 rows with 8 trees in each row. How many trees?	**64**	8 × 8 = 64 trees.

ID	Question	Answer	Why (explain to student)
bb4_s2	Set 3: Find the mistake. Mia wrote: 6 × 8 = 6 × 4 + 6 × 4 = 24 + 24 = 46. Which part is wrong?	**24 + 24 is 48, not 46.**	Breaking 8 into 4 + 4 works, and 6 × 4 = 24. But 24 + 24 = 48, so 6 × 8 = 48.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	4 rows of 5 jars	4 × 5 = 20 jars (5 × 4 = 20 also accepted).
2	Array of 3 rows of 6 trees	Array shows 3 rows with 6 in each row; 3 × 6 = 18 trees.
3	Same product as 6 × 8	8 × 6 (both equal 48).
4	Skip-count by 6s	24, 30, 36
5	9 × 1 and 7 × 0	9 × 1 = 9; 7 × 0 = 0

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Break-apart 7 × 8	7 × 8 = 7 × 5 + 7 × 3 = 35 + 21 = 56
2	9 crates of 6 oranges	9 × 6 = 54 oranges; turn-around 6 × 9 = 54.
3	7 rows of 9 bins, 8 unloaded	Step 1: 7 × 9 = 63. Step 2: 63 − 8 = 55 bins.
4	Mistake in 9 × 4 = 63	He switched the digits. In the 9s the tens digit is one less than 4, so 9 × 4 = 36 (3 + 6 = 9).
5	Vocabulary match	factor: a number you multiply; product: the answer to a multiplication problem; turn-around property: 6 × 8 = 8 × 6; zero property: any number times 0 equals 0.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	8 rows of 9 trees plus 2 more rows of 9	90 trees. Way 1: 8 × 9 = 72, 2 × 9 = 18, 72 + 18 = 90. Way 2: 8 + 2 = 10 rows, 10 × 9 = 90.